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FOB 101 - Assignment 3

1. The balanced equation of decarboxylation of acetic acid to CO_2 is

$$2\ CH_3COOH + O_2 + 4H^+ \longrightarrow 2\ CO_2 + 2H_2O$$

2 oxidations => 2 CH_3COOH from +3 to +4
2 decarboxylations => 2 CH_3COOH forming CO_2

The complete reaction is

$$CH_3CO\text{-}S\text{-}CoA + 2O_2 + 4H^+ \rightleftharpoons CH_3COOH + HS\text{-}CoA + 2H_2O$$

For balancing

$CH_3COOH \rightleftharpoons CO_2 + 2H^+ + 2e^-$ (oxidation half)

$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ (reduction half)

2. The correct order is
T S R P Q U

3. Fumarate and FADH2 are OXIDISING AGENTS

Succinate and FAD are REDUCING AGENTS

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4. a) 20 , 40 , 20

b) ATP (Adenosine triphosphate) is more reactive because of its structure and high energy bonds.

- Bulky with high steric hindrance
- OHs attatched to phosphate groups create repulsion.
- Phosphate groups are attatched by high energy phosphate bonds.
- Phosphate groups can be easily hydrolysed.

5. a) Sunlight (Light energy)

b) Light reaction

c) Dark reaction (Calvin cycle)

d) O_2 is released

e) ~~Electron transport chain~~
e) Photosystem

f) NADPH

g) ATP

h) GSP